Tracing Morph Origins in Czech: A Computational Approach to Morph-Level Etymology

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Abstract

This work introduces a new machine learning task at the intersection of morphology and etymology, predicting the origin of individual morphs (minimal meaning-bearing units) in Czech words. Given morphologically segmented Czech sentences, the aim is to determine for each morph whether it is native or borrowed and, if it is borrowed, through which languages it entered Czech. Although some linguists have examined etymology at the level of individual morphs rather than whole words (Arkadiev et al., 2015), to our knowledge, no prior computational study has addressed this task at the morph level.

We created a manually annotated dataset of 300 Czech sentences comprising around 10,000 morphs with morph-level etymology labels, and evaluated several machine learning approaches using character-based and structural features. Our best lightweight system is a feed-forward neural network with a single hidden layer, trained on data augmented with entries from an etymological dictionary, reaching 96.5 % F1 on the test set. We also designed and evaluated multiple in-context prompting variants for large language models. In the setting where the full training set is provided in the prompt, the best model, `Claude Opus 4.5`, achieved 97.8 % F1.

The present work is a direct continuation and substantial extension of the bachelor thesis *Identification of Morpheme Origin* (Papáček, 2025). The thesis covers the theoretical background in greater detail, while this version focuses on substantial experiments with several large language models and on an application of the learned model to a diachronic Czech legal corpus. The author's main contribution lies in the creation and manual annotation of the dataset, implementation of the experimental pipeline, execution of the experiments, and analysis of the results. The work was developed under the supervisor's guidance, particularly in the design of the experiments and in shaping the final form of the text.

A version of this work was accepted for presentation at the *4th Workshop on Language Technologies for Historical and Ancient Languages*, co-located with LREC 2026, a conference ranked B in the CORE ranking. The code, prompts, and dataset are released as open source at `https://github.com/ampapacek/MorphemeOrigin`.

Keywords: Etymology, Morphology, LLMs, Machine Learning

1. Introduction

Borrowing is one of the main forces driving language change (Grant, 2020). English, for example, owes much of its vocabulary to French and, through it, to Latin. Latin itself borrowed extensively from Greek (Durkin, 2014). What gets borrowed is not always uniform at the word level: borrowed roots often combine with native morphology, and native roots can take borrowed affixes.

Modern European languages therefore contain many hybrid words whose morphs come from different sources. For instance, *antiviruses* combines the Greek-derived prefix *anti-* with the Latin root *virus* and the English inflectional ending *-es*.

Morphs are the building blocks of words, yet most existing etymological resources stop at the word level. Studying morph origins offers a much finer view of language history and contact. Linguistic research has explored morph borrowing in works such as Arkadiev et al. (2015) and Seifart (2015). However, so far, existing computational approaches have focused on word-level etymology rather than morph-level analysis.

Automatically predicting morph origins is difficult: the space of possible morphs is large, and many morph surface forms are short and ambiguous, meaning the same form can occur in different words with different etymological origins. This is also difficult to model computationally because much of the relevant knowledge is implicit in linguistic expertise, and existing etymological resources are mostly word-level rather than morph-level.

We frame morph-level etymology prediction as a supervised learning task. Given a segmented sentence, the goal is to decide for each morph whether it is native or borrowed and, if borrowed, which languages are involved in its borrowing path.

To enable this, we manually annotated almost 10,000 morphs in 300 sentences from the SIGMORPHON 2022 shared task (Batsuren et al., 2022) with morph-level etymological labels.

Figure 1: Simplified illustration of morph-level origin prediction for the phrase “antivirový program” (*antivirus program*). For readability, the figure shows only one origin label per morph, represented by flags (Greek, Latin, and Czech).

We compare several baselines, including a majority-label baseline, memorization of morph–origin pairs, and rule-based methods using the Czech Etymological Lexicon (Rejzek et al., 2025). For the supervised approach, we extract character-based and structural features (character n-grams, morph type, and position within the word) and train several classifiers: logistic regression (LR), support vector machines (SVM), multilayer perceptrons (MLP), and a bidirectional gated recurrent unit (GRU) model over morph sequences. We also test morph and word embeddings and a semi-supervised self-training setup.

In addition, we evaluate predictions from several state-of-the-art LLMs under different in-context training settings.

Although the general approach is language-independent, our experiments focus on Czech, where suitable etymological resources are available, primarily the Czech Etymological Dictionary (Rejzek, 2019). We believe that the proposed methodology should in principle be transferable to other languages, given the availability of morph-level etymological annotations.

This article builds on the bachelor thesis *Identification of Morpheme Origin* (Papáček, 2025), which presents the theoretical background and model setup in greater detail. Compared with the thesis, the present version condenses the material into a conference-paper format, adds experiments with several recent large language models, and includes an appendix applying the model to a diachronic Czech legal corpus. The work also builds on the author’s involvement in the development of relevant language resources, including the Czech Etymological Lexicon 1.0 (Rejzek et al., 2025), UniSegments 1.5 (John et al., 2026), and DeriNet 2.3 (Olbrich et al., 2025).

A version of this article was accepted for presentation at the *4th Workshop on Language Technologies for Historical and Ancient Languages* (LT4HALA Organizers, 2026), co-located with LREC 2026 in Spain. We thank the anonymous reviewers for their constructive feedback, which helped improve the final version of the paper.

The rest of the paper is organized as follows. Section 2 introduces key concepts from morphology and etymology. Section 3 reviews related research. Section 4 describes the data and annotation process, and Section 5 outlines the machine learning approaches. Section 6 explains the evaluation setup, and Section 7 reports and analyses the experimental results.

2. Background Theory

One of the central tasks of morphology is to investigate the internal structure of words, often by segmenting them into the smallest meaning-bearing units, known as **morphs**.[1] For example, the English word *disrespectfulness* can be decomposed into *dis-* + *re-* + *spect* + *-ful* + *-ness*. Each morph carries lexical or grammatical meaning.

2.1. Basic Terminology

The distinction between *morpheme* and *morph* is notoriously vague (Haspelmath, 2020). Haspelmath observes that the term *morpheme* is used in several conflicting senses, ranging from concrete minimal forms to abstract grammatical units. When we work with segmented word forms, we refer to concrete realizations such as *-s* or *-es*, rather than abstract categories like *plural*. In that case, we use the term *morph* in the sense defined by Haspelmath (2020): a morph is a minimal linguistic form.

The **root** is the core morph of a word and carries its main lexical meaning. Morphs that must attach to a root are called **affixes**, which modify either lexical meaning or grammatical properties. **Prefixes** attach before the root, while **suffixes** follow it. An **ending** is a specific type of suffix that marks grammatical features such as tense, person, or number. An **interfix** appears between roots.

Affixes are commonly classified as either **inflectional** or **derivational**. **Inflection** modifies a word’s grammatical form without changing its lexical meaning, whereas **derivation** creates a new lexeme, often with a different meaning or part of speech (Aronoff and Fudeman, 2011).

2.2. Etymology

Etymology studies how words originate and evolve over time, and whether they are inherited or borrowed from other languages. Morph-level etymology examines this at the level of individual morphs,

[1] In current theoretical debates, word-based approaches to morphology are gaining prominence (Blevins, 2016). These approaches treat words as the primary units of morphological analysis, focusing on relations between whole words, and regard morphs as secondary abstractions derived from them. However, we believe that modeling the origins of individual morphs, as pursued in this paper, is compatible with viewing morphs as secondary abstractions.

offering finer-grained insights, especially for hybrid words whose parts come from different sources.

In much of the language contact literature, the common assumption is that affixes are usually borrowed only indirectly. That is, words containing a given affix are first borrowed as wholes from another language, and only later are these words reanalysed, and the recurring affix generalized to native stems.

For example, the suffix *-ism* (from Greek, via Latin and French) entered English through borrowed words such as *baptism*, *criticism*, and *organism*, and was subsequently extended to native bases in formations like *snobbism* or *veganism*.

There are, however, also cases where affixes have been borrowed directly, without the mediation of complex loanwords. Although such cases are rare (Seifart, 2015), examining etymology at the level of individual morphs can still provide valuable insights, even when most affixes have entered a language through indirect borrowing.

Borrowing can also occur through a chain of intermediary languages. For example, the English word *school* entered Middle English as *scole* from Old English *scol*, from Proto-West Germanic **skolu*, from Late Latin *schola*, from Ancient Greek *skhole*, and probably from Proto-Indo-European (Oxford English Dictionary, 2025).

We should note that our assumption that historical pathways of words can be captured rigorously as plain sequences of languages is an operational simplification. The etymology of a word is not always straightforward; for various reasons, it may be difficult or impossible to determine the exact trajectory by which it entered the language or the intermediate languages through which it passed.

This is particularly evident in the case of calques. For example, Czech *časopis* ('magazine') was formed as a translation of German *Zeitschrift*, and *mrakodrap* ('skyscraper') as a translation of English *skyscraper*. Here, what is borrowed is primarily the structural pattern and meaning, while the morph material itself is native. As a result, it is not entirely clear whether such words should be classified as native or borrowed: from the perspective of whole words they may be viewed as borrowings, but from the perspective of individual morphs their components remain native.

3. Related Work

To our knowledge, no computational linguistics research specifically addresses identifying the etymological origin of individual morphs on a larger scale. While there is some research on affix borrowing and a few tools exist for determining the etymology of entire words, no approach combines these perspectives at the morph-level.

3.1. Linguistic Research on Morph Borrowing

Linguistic research has examined how morphology spreads across languages through language contact and how such processes can be distinguished from ordinary inheritance. Johanson and Robbeets (2013) draw a key distinction between shared affixes that are *copies*—replicated through contact—and those that are *cognates* inherited from a common ancestor, framing borrowing as an active process of *code-copying*. Seifart (2015) further distinguishes between *direct* borrowing, based on speakers' knowledge of the donor language, and *indirect* borrowing, where affixes spread through complex loanwords that later become productive in the recipient language.

Gardani (2008) shows that even inflectional morphs—traditionally viewed as resistant to borrowing—can transfer when language contact is intense and structural compatibility is high. Broader perspectives are provided by Arkadiev et al. (2015), which focuses on the borrowing of affixes and shows that, although less common than lexical borrowing, derivational affixes and some basic grammatical endings can also be transferred between languages. Together, these works form a theoretical basis for our approach to studying morph-level etymology computationally.

3.2. Computational Etymology

Building on this linguistic background, recent computational studies have begun to model etymological relationships automatically, though still mostly at the word level. Wu and Yarowsky (2020) predict both the type of etymological relationship (e.g., borrowing or inheritance) and the parent language from which a word originated. Their approach shows that computational modelling of etymology is both feasible and informative, though it does not address the internal morphological structure of words.

The work by Kovácsová (2025) explores methods for training models to automatically identify loanwords across multiple languages. Our research extends this line of inquiry to the morph-level, providing a finer-grained view of how native and borrowed elements combine within words. It also advances beyond binary or single-language classification by predicting the set of languages involved in the morph's etymological origin.

3.3. Resources with Morph-Level Etymological Information

Despite the growing interest in computational etymology, few resources provide morph-level annotation. Wiktionary (2025) includes occasional etymologies for affixes and bound morphs, but coverage is sparse and inconsistent. Similarly, the *Czech Etymological Dictionary* (Rejzek, 2019) contains etymological information for several dozen affixes but focuses primarily on whole words.

Another relevant source is the open dataset by Goldberg (2019), which lists morphs with basic linguistic information. Around 1,500 entries include etymological notes—mostly for technical affixes such as *acet*-, *agri*-, *exo*-, or -*ant*—derived mainly from Greek or Latin. However, the dataset lacks detailed source references and provides only minimal etymological depth.

Overall, these resources offer isolated examples of morph-level etymology but lack the consistency and structure required for systematic computational modelling. Our work aims to fill this gap by developing a comprehensive, linguistically grounded dataset of Czech morphs with etymological labels.

4. Data

This section describes the linguistic resources, annotation process, and datasets used for training and evaluating our models.

4.1. Czech Etymological Dictionary

The latest Czech Etymological Dictionary (Rejzek, 2019) contains over 11,000 main entries and around 21,000 derived forms.

Its digital version, the Czech Etymological Lexicon (**CzEtyL**) (Rejzek et al., 2025) reformats this information into a structured, machine-readable format. Each entry specifies whether a word is native or borrowed and, for borrowed words, lists the sequence of languages through which it passed before entering Czech.

Example entry:

```
architekt    deu,lat,ell Loan
```

The word *architekt* ("architect") comes from Greek, entering Czech via Latin and German.

We extended CzEtyL using derivational relations from DeriNet (Olbrich et al., 2025), a large Czech lexical network linking derived words with their bases. For each CzEtyL entry, we added all lexemes from the same DeriNet tree and assigned them the same etymological label, expanding the coverage from about 10,500 to over 500,000 annotated words. This expansion is only an approximation, it assumes that lexemes in the same derivational tree preserve the etymology of the root. This improves coverage but may introduce some noise.

Using DeriNet segmentation together with the original and expanded CzEtyL, we also derived approximate root and affix dictionaries: root labels were obtained by assigning the word-level etymology to the root morph, while affix labels come from the affix entries explicitly listed in CzEtyL. These dictionaries are used both in the rule-based baselines and as additional training material for the supervised models.

4.2. Our Annotation

We used the Czech part of the morpheme segmentation dataset from (SIGMORPHON, 2022), which provides sentences segmented into morphs. Each morph was automatically classified as *root*, *derivational*, or *inflectional affix* using the model of John (2024), and then manually annotated with its sequence of language origins based mainly on Rejzek (2019) and Wiktionary (2025). Because etymological sources usually operate at the word level rather than the morph level, morph-level annotation has no widely established standard guidelines, and is therefore hard and sometimes ambiguous.

Example: *Šetřete peníze netelefonujte faxujte.*
Translation: *Save money, don't call, fax instead.*

- **Šetřete** (*save*)
  - *Šetř* — `R` — `ces`
  - *e* — `D` — `ces`
  - *te* — `I` — `ces`
- **peníze** (*money*)
  - *peníz* — `R` — `deu, lat`
  - *e* — `I` — `ces`
- **netelefonujte** (*don't call*)
  - *ne* — `D` — `ces`
  - *tele* — `R` — `ell`
  - *fon* — `R` — `ell`
  - *uj* — `D` — `ces`
  - *te* — `I` — `ces`
- **faxujte** (*fax*)
  - *fax* — `R` — `eng, lat`
  - *uj* — `D` — `ces`
  - *te* — `I` — `ces`

The annotation distinguishes three morph types: roots (`R`), derivational (`D`), and inflectional affixes (`I`). Language origins use ISO 639-3 codes such as `ces` (Czech), `deu` (German), `lat` (Latin), and `ell` (Greek).

The dataset was annotated by two native Czech speakers: a university student with a Bachelor's background in NLP and etymology, and an IT university student without prior formal linguistic training. Both annotators received detailed guidelines and a short calibration on example sentences, as well as access to resources such as the Czech

Dataset	Sentences	Words	Morphs
Train	200	2,904	7,212
Dev	50	632	1,501
Test	50	643	1,532

Table 1: Size of the annotated dataset used for training, development, and testing

Etymological Dictionary and Wiktionary (2025). Although neither is a professional linguist in this field, their work provides a useful reference point for human performance on this task.

Cohen's kappa was 0.82, and the exact match rate reached 95.9 %, indicating strong agreement. Remaining differences mostly stemmed from varying annotation granularity (e.g., `gmh` vs. `deu`) or inconsistencies in etymological sources.

4.3. Training and Evaluation Data

The dataset was then split into training, development, and test sets. Table 1 shows their sizes in terms of sentences, words, and morphs. The counts in Table 1 include all morphs present in the annotated files, including items without an etymology label, such as some numerals. Sequence-frequency statistics reported below are computed only over morphs with non-empty etymological annotation.

We drew the training set from the Czech SIGMORPHON (2022) Shared Task training data (200 sentences). We derived the development and test sets from the 500-sentence development file by selecting every 10th sentence for testing and every 10th sentence for development (offset by 1), leaving the remaining sentences unused for possible future work. Dataset size is limited by the cost of manual annotation, which requires linguistic expertise or substantial background research.

The training data contains 7,073 morphs with non-empty etymological labels, but most fall into just a few common etymological classes (Table 2). The etymology column lists language codes using the ISO 639-3 standard. As expected, the majority of morphs are of Czech (native) origin. Since the training set covers only 200 sentences, it should be seen as a small sample and not a fully representative picture of Czech in general.

5. Proposed approaches to Morph-Level Etymology Prediction

5.1. Baselines

To evaluate performance, we define several baselines to establish a lower bound on expected results and assess how much the model improves beyond simple heuristics. These include always predicting Czech, a memorization approach, and methods using *CzEtyL*.

Rank	Etymology	Count	% of total
1	ces	6,231	88.1
2	lat	281	4.0
3	ell	105	1.5
4	deu,lat	84	1.2
5	eng,lat	41	0.6
6	lat,ell	38	0.5
7	ita,deu	38	0.5
8	eng	36	0.5
9	deu,lat,ell	29	0.4
10	fra,lat	26	0.4
–	**Rest**	164	2.3

Table 2: Top 10 most frequent etymological sequences in the training data.

Always-native This baseline predicts all morphs as native (`ces`). It serves as a simple reference point and performs well when native morphs dominate the dataset.

Memorization We store all training morphs and predict their most frequent etymology label. Unseen morphs default to the majority class (`ces`). This baseline fails on ambiguous forms whose origin depends on context. For example, the suffix *-um* in *muzeum* ('museum') is of Latin origin, while the *um* in *rozum* ('reason, mind') is native Czech, derived from the verb *umět* ('to be able to').

A related problem appears in homonymous words such as *kolej*: one sense means 'railway' and is related to the native root *kolo* ('wheel'), while the other means 'college' and ultimately goes back to Latin *collēgium* (Rejzek, 2019).

Word lemmatization Each word is lemmatized with *MorphoDiTa* (Straková et al., 2014) and the lemma is looked up in the extended *CzEtyL* (4.1). The retrieved etymology is assigned to the root morph(s). Non-root morphs are matched against a *CzEtyL*-derived list of borrowed affixes; unseen affixes and roots default to `ces`, and inflectional endings are always predicted as `ces`.

5.2. Feature-based Classification

To predict the etymological origin of each morph, we extract features from annotated data that capture the characteristics of the morph and its context. These features serve as input to the classification model. We use character n-grams, morphological classification, position within the word, and embeddings, among others. They serve as input to the classification models used in our experiments.

We evaluate the following standard classifiers,[2] Logistic Regression (LR), Support Vector Machines (SVM), and Multi-Layer Perceptrons (MLP).

Character n-grams Character n-grams capture language-specific letter patterns and help the model associate certain combinations with particular etymological origins. We use 1-grams and 2-grams, which are appropriate given the average morph length of about 2.2 characters.

Morph Types Each morph is categorized as *Root*, *Derivational affix*, or *Inflectional affix*, encoded as a one-hot vector. Positional classification is also used, labelling morphs as *Root*, *Prefix*, *Suffix*, or *Interfix* depending on their position relative to the root(s) in a word. This classification is obtained using a model from John (2024).

Vowel Start and End This feature encodes whether a morph starts or ends with a vowel, using two binary values. Some languages favour vowel-initial or vowel-final patterns, so even such a simple feature can capture useful language-specific tendencies.

It becomes especially informative when combined with morph type. Many Proto-Indo-European roots, for example, follow a consonant-vowel-consonant (CVC) structure (Gamkrelidze and Ivanov, 1995). Affixes, by contrast, often attach more smoothly when starting or ending with a vowel. As a result, prefixes frequently end in vowels, while suffixes often begin with them.

Impact of Embeddings We use FastText embeddings (Bojanowski et al., 2017), which represent each word as the sum of its character n-gram embeddings. This method enables generating vectors even for morphs that are not stand-alone words. We use Czech FastText embeddings trained on Wikipedia and Common Crawl data (Grave et al., 2018).

We experiment with two variants: using embeddings directly for individual morphs, and using embeddings of the entire words in which the morphs occur.

5.2.1. Representation of Target Classes

We model target classes using two strategies:

- **Whole-sequence classification:** Treats each language sequence as a single label, predicting only sequences seen in training.
- **Multi-label classification:** Treats each language independently, allowing prediction of unseen combinations by training a binary classifier per language.

5.2.2. Data Augmentation

Beyond the original training set, we experiment with expanding the training material using two additional sources: data from the CzEtyL lexicon and automatically self-labeled data produced by our own model.

Use of Data from CzEtyL We extracted a dictionary of roots and affixes from *CzEtyL* with their probable etymological origins to serve as additional training data for the supervised models. The extracted root and affix dictionaries together contain roughly 12,000 morphs, covering about 400 distinct language sequences and 67 individual languages. These dictionaries are not perfect, as the root dictionary was constructed by assigning each root the most frequent etymology found among the words containing it, which can occasionally introduce errors.

Self-Training Self-training is a semi-supervised approach where a model is first trained on a manually annotated dataset, then used to label a larger unlabelled corpus. The newly labelled examples are added to the training set to improve the model. This is useful when additional annotated data is costly or difficult to obtain.

In this work, we applied self-training using a portion of the Prague Dependency Treebank (PDT) (Hajič et al., 2020). The selected subset contains about 88,000 sentences and approximately one million running words. The corpus was segmented automatically using a tool developed on top of the UniSegments project (John et al., 2026). Before processing, the text was normalized by lowercasing and removing punctuation, numbers, and special characters.

5.3. GRU Sequence Model

Besides the feature-based classifiers above, we also experiment with a neural sequence model based on bidirectional gated recurrent units (GRUs). Each morph is first encoded from its characters using a bidirectional GRU. This representation is then concatenated with embeddings of the morph type and morph position, the binary vowel-start/end features, and optionally FastText embeddings of the morph itself and of the full word in which it occurs.

[2] We use the implementations from `https://scikit-learn.org`.

A second bidirectional GRU then processes the sequence of morphs and predicts one etymological label per morph. We tested two scopes: processing morphs within each word, and processing all morphs in a full sentence. In practice, the word-level setup worked clearly better, so we report word-level GRU results in the main comparison.

5.4. In-context LLM Setup

We evaluated several LLMs through prompting only, without fine-tuning. Since some settings include labelled training examples directly in the prompt, we refer to this as an in-context LLM setup. The models were asked to label Czech morphs with ISO 639-3 origin codes, including intermediary donor languages when relevant, while preserving the input structure and morph-type tags. We tested `GPT-5.2`, `GPT-5-mini`, `GPT-4o-mini`, `Claude-Opus-4.5`, `Gemini-3-Pro`, and `Deepseek-Chat-v3.1`.

We tested multiple prompt variants and selected the one that produced the most consistent, automatically parseable outputs. The final instruction prompt defines the required output format, annotation rules, and the meaning of ISO 639-3 language codes and morph-type tags.

All models were queried via the OpenRouter API[3]. For each in-context setting, the same fixed set of training examples was used for all models and all evaluated batches. The tested context sizes were 0, 50, 200, 2,000, and the full training set (∼7,000 morphs). The target development or test data was split into batches of roughly 350 morphs, rounded to full sentences. Each request therefore contained the instruction prompt, the selected in-context training examples, and one target batch. Each batch was processed in a fresh chat/session, with no conversation history from previous batches.

The full prompt, API scripts, and data are released with the code repository at https://github.com/ampapacek/MorphemeOrigin. We used temperature $= 0$ and otherwise default API settings to support reproducibility.

6. Evaluation Methodology

The task is to predict, for each morph, the sequence of languages through which it entered Czech, including any intermediate stages of borrowing. This sequence represents a historical borrowing path when such a path is known.

In the experiments, however, we do not evaluate the order of these language codes. For modelling and evaluation, we treat each annotation as a set of source languages, so for example `deu, lat` and `lat, deu` are counted as the same prediction. This choice lets us focus on identifying which languages participate in the origin of a morph. In some cases, chronological order may be approximated using rule-based methods based on known contact periods between languages, or by a separate model, but we leave systematic reconstruction of the ordered path for future work.

We use an example-based F1-score to evaluate prediction quality. For each morph occurrence with a non-empty gold etymological annotation, we compare the predicted set of languages P with the gold set G and compute

$$F1 = \frac{2|P \cap G|}{|P| + |G|}.$$

The main score is the average of this value over all evaluated morph occurrences. In single-label settings, where each prediction contains only one etymological class, this behaves like exact-match scoring at the morph level; in the multi-label setting, it also gives partial credit when only some languages in the borrowing path are predicted correctly.

To account for class imbalance between native and borrowed morphs, we also compute separate F1-scores for these two groups (**Bor.**, **Nat.**). These scores provide a clearer view of model performance, because borrowed morphs are generally harder to classify due to diverse sources and complex borrowing paths.

Many morphs appear multiple times in the dataset, especially inflectional endings. In the training set, there are 7,212 total morph instances, with 1,028 distinct morphs. The top 10 morphs account for 1,969 occurrences (27 %); the top 20 cover 2,873 (40 %); and the top 50 make up 4,007 occurrences (56 %)—over half the dataset.

To reduce bias from frequent morphs, we also report an additional metric (**Dst.**): for each distinct morph, we first compute its average F1-score across all its occurrences and then take the mean of these per-morph scores to obtain the final value.

We also report the relative error reduction (**RER**) of the F1 score over the all-native baseline to better illustrate each model's improvement.

7. Results

We use the development set for model selection and report final numbers on the held-out test set. For the supervised models, we tuned hyperparameters with a grid search over feature and classifier settings. For LLMs, we compared a small set of in-context prompting configurations, varying the prompt itself, in-context training size, and batch size.

[3] We enabled OpenRouter's "do not train on my data" setting for all requests.

7.1. Learning models: base results

Table 3 reports development-set performance of the supervised classifiers. All models use *single-label* prediction, treating each full etymological sequence as one class. The SVM uses an RBF kernel, and in the MLP model names the number indicates the hidden-layer size (number of neurons). For the GRU, we report the plain word-level setup without external embeddings or *CzEtyL*-derived augmentation.

Model	F1	RER	Nat.	Bor.	Dst.
LogReg	94.0	39.7	98.7	51.4	86.2
SVM	94.7	46.5	**99.5**	50.7	88.1
GRU	95.1	50.3	98.9	60.0	89.3
MLP30	95.5	54.5	98.9	64.1	90.1
MLP100	**95.6**	**55.5**	99.0	**64.4**	**90.4**
MLP300	95.5	54.3	99.3	60.5	90.2

Table 3: Performance on the development set. RER = relative error reduction over the All-native baseline; Nat. / Bor. = F1 on native / borrowed morphs; Dst. = average F1 over distinct morphs.

Both logistic regression and SVM struggled on borrowed morphs, although SVM achieved the best score on native ones. The plain GRU already outperformed LR and SVM and was competitive with the simpler MLP variants, especially on borrowed morphs. All MLP variants performed better overall, and results were very similar across hidden-layer sizes. In our experiments, adding extra hidden layers did not yield a consistent improvement.

7.2. GRU-based sequence model

While Table 3 shows the plain GRU setup, we also tested several representative GRU variants differing in sequence scope and in whether FastText embeddings were used for the morph, the containing word, or both. Table 4 summarizes the main development-set patterns using compact code names explained below the table. The word-level setting was clearly better than processing full sentences, and adding FastText embeddings brought only moderate gains unless combined with additional training data from *CzEtyL*. The best development configuration combined word-level processing with both morph and word embeddings and with *CzEtyL*-derived augmentation; this variant reached 96.2 % F1 on the development set, with 68.4 % F1 on borrowed morphs and 91.4 % on the distinct-morph metric.

Codes: ‘S’ = sentence-level scope, ‘W’ = word-level scope, ‘0’ = no embeddings, ‘W’ = word FastText, ‘M’ = morph FastText, ‘B’ = both word and morph FastText, ‘+C’ = additional *CzEtyL*-derived training data.

Code	F1	RER	Nat.	Bor.	Dst.
S0	90.1	0.0	**100.0**	0.0	79.4
SB	92.5	24.0	99.6	27.5	83.5
W0	95.1	50.3	98.9	60.0	89.3
WW	95.3	52.4	98.9	62.1	89.8
WM	95.3	52.6	98.9	63.0	89.9
WB	95.6	55.3	99.3	61.5	90.2
WB+C	**96.2**	**61.5**	99.2	**68.4**	**91.4**

Table 4: Representative development-set GRU configurations. FastText embeddings are used either for the morph, the full word, or both; *CzEtyL* indicates training-data augmentation with dictionary-derived roots and affixes.

Overall, the development-set trend suggests that the GRU is a viable alternative to the simpler feed-forward models, but not a clearly superior one in our setting. One possible reason is that the relevant sequences are usually very short: most words contain only a small number of morphs, so the advantage of a recurrent model over a simpler MLP may remain limited in this task. Another likely factor is data size. Sequence models with learned embeddings typically benefit from more labeled data than we had available, so the GRU may be somewhat underutilized in the present dataset size.

7.3. Embeddings and Multi-label Setup

We evaluated several ways of integrating embeddings into the model, using representations of the morph itself, the word in which it appears, and their combination. The experiments were run with multiple model configurations. Overall, results were very similar across these settings, suggesting that embeddings did not lead to substantial improvement. Table 5 reports results for MLP30 on the development set.

We also tested a multi-label setup, where each language is predicted independently. Instead of selecting a single etymological sequence, the model predicts which languages belong to the etymological path of a given morph. This increases flexibility and allows combinations not seen in training. In practice, the setup trains one small classifier per language, effectively increasing total capacity.

In Table 5, we compare multi-label models trained on the base data with models extended by roots and affixes from *CzEtyL*. In the model names, **M** denotes the multi-label setup and **C** marks the inclusion of *CzEtyL* data. Multi-label prediction alone did not improve performance on the base training set, but it brought clear gains when combined with *CzEtyL*-derived data. Increasing the hidden-layer size brought no additional benefit.

Setting	F1	RER	Nat.	Bor.	Dst.
Effect of embeddings - MLP30 model					
No Emb.	95.5	54.5	**98.9**	64.1	90.1
Word	95.0	49.5	98.7	61.2	89.2
Morph	95.1	50.8	**98.9**	60.4	89.4
Word+Morph	95.3	52.5	98.7	64.2	89.4
Multi-label MLPs					
Without CzEtyL data					
MLP30 M	95.3	52.8	98.8	64.1	89.7
MLP100 M	95.1	51.1	98.8	61.6	89.3
MLP300 M	95.0	49.8	**98.9**	59.6	89.1
With CzEtyL data					
MLP30 MC	**95.8**	**58.2**	98.8	69.5	**90.9**
MLP100 MC	95.6	56.1	**98.9**	66.0	90.7
MLP300 MC	**95.8**	57.3	98.4	**71.6**	90.6

Table 5: Development-set results: (i) effect of embeddings for MLP30, and (ii) multi-label MLP variants with and without additional CzEtyL-derived training data.

#Morphs	F1	RER	Nat.	Bor.	Dst.
GPT-4o-mini					
0	90.3	2.0	99.9	2.6	80.0
50	90.3	2.6	99.9	3.3	80.2
200	90.2	1.5	99.9	2.2	79.9
2,000	90.1	0.1	99.9	1.5	79.4
Full	90.4	3.2	99.9	3.9	80.1
GPT-5-mini					
0	94.4	43.6	98.5	57.4	90.0
50	95.5	54.6	99.5	59.4	91.3
200	94.7	47.0	99.5	51.9	89.5
2,000	95.7	56.7	99.5	60.9	90.9
Full	96.4	63.6	99.6	67.0	92.5
GPT-5.2					
0	94.7	47.1	98.6	59.5	89.6
50	95.5	54.2	98.9	64.5	90.6
200	95.6	55.3	98.8	66.3	90.8
2,000	95.3	52.6	98.8	63.6	90.3
Full	96.8	67.4	99.1	**75.7**	92.8
Claude-Opus-4.5					
0	95.7	56.4	99.3	62.6	90.9
50	95.1	50.7	99.3	56.9	90.2
200	95.4	53.6	99.3	59.8	90.6
2,000	96.0	60.3	99.4	65.8	91.8
Full	**97.0**	**69.4**	99.4	74.9	**93.6**

Table 6: Development-set in-context LLM results with different numbers of training morphs included in the prompt. *Full* corresponds to the whole training set (∼7,000 morphs).

7.4. In-context LLM results on the development set

Table 6 shows how performance changes with the number of training morphs included in the prompt. A consistent pattern across models is that native-morph performance is near ceiling, so most differences come from how well the models handle borrowed morphs.

`GPT-4o-mini` stays close to the all-native baseline because it rarely identifies borrowed morphs correctly, even when given many in-context examples. One possible explanation is that the smaller capacity of this model makes it less reliable on the subtle decisions required for short and ambiguous morph strings, so it often falls back to the dominant native label.

The effect of adding in-context examples is not entirely consistent. For some models, a small number of in-context examples does not improve performance and may even make it slightly worse. Still, the full-training-set setting gives the best development-set F1 for `GPT-5-mini`, `GPT-5.2`, and `Claude-Opus-4.5`. The gain is especially clear on borrowed morphs. For instance, `GPT-5.2` rises from 59.5 % F1 in the zero-shot setting to 75.7 % F1 when the full training set is included in context.

We also tested `Gemini-3-Pro` and `Deepseek-Chat-v3.1` they were both close to the `GPT-5-mini` but below `Claude-Opus-4.5` and `GPT-5.2`.

Model	F1	RER	Nat.	Bor.	Dst.
Baselines					
All native	89.9	–	**100.0**	0.0	77.1
Memor.	94.4	44.9	99.3	51.5	86.3
Lemmatiz.	94.8	48.1	98.6	60.7	90.0
Learning models					
MLP30	95.2	51.6	98.6	65.2	88.3
MLP100 M	95.8	57.4	99.1	66.2	90.1
MLP300 MC	96.2	62.0	98.9	72.7	90.8
MLP30MC	96.5	65.6	99.6	69.6	91.5
GRU (best)	96.4	64.7	99.3	71.4	91.0
Self-train	96.1	60.8	98.6	72.8	91.0
In-context LLMs					
Cld.-Opus4.5	**97.8**	**77.7**	99.7	**80.4**	**94.7**
DeepSk-v3.1	96.5	65.6	99.6	69.0	91.9
Gemini-3pro	96.5	65.8	99.7	68.4	91.8
GPT-4o-mini	90.9	9.4	99.9	10.1	79.4
GPT-5-mini	96.4	64.0	99.8	66.0	92.5
GPT-5.2	97.6	76.6	99.8	78.6	94.4

Table 7: Final results on the test set. In-context LLMs use the full training set in the prompt.

7.5. Results on Test Set

We now evaluate all models on the test set, including the baselines, learning-based approaches, and the in-context LLM setup with the full training set provided in the prompt. Table 7 summarizes the results.

7.6. Discussion of Results

In-context LLMs With the full training set provided in the prompt, the LLMs reach strong performance overall (Table 7). `Claude-Opus-4.5` is the top performer in this setting, closely followed by `GPT-5.2`. `GPT-5-mini` and the other non-OpenAI models (`Gemini-3-Pro`, `Deepseek-Chat-v3.1`) form a second tier with similar results. In contrast, `GPT-4o-mini` lags far behind, and the gap is driven mainly by weak performance on borrowed morphs rather than on native ones, which are near ceiling for all models.

Baselines The trivial *All native* model achieved an F1 score of 89.9 %, which also reflects the proportion of native morphs in the dataset. The lemmatization approach, which draws on etymological data from CzEtyL, cut this error by nearly half. It outperformed the simpler memorization baseline.

Learning models The trends from the development set carried over to the test set. Using *CzEtyL*-derived training data and the multi-label formulation both helped, with the best supervised variants reaching around 96.5 % F1 (e.g., MLP30MC: 96.5).

The GRU sequence model was also competitive. Its best variant reached 96.4 % F1 on the test set, slightly below MLP30MC overall, but somewhat stronger on borrowed morphs (71.4 vs. 69.6). However, its overall advantage over the simpler MLPs remained limited. We suspect this is partly because morph sequences within words are short, and partly because the GRU likely needs more labeled data to fully benefit from its higher-capacity sequential representation.

In the self-training setup (5.2.2), we used the best development-set model (MLP30MC) to label the large corpus and then retrained the model on these pseudo-labels. The resulting model reached a very similar overall score (96.1 % F1). It corrected some morphs, but also introduced new errors, so in the end it did not bring a clear improvement over the fully supervised approach. On borrowed morphs, its performance was slightly better and among the strongest of the learning-based models. However, this came with worse results on native morphs.

Performance by Morph Type Since borrowing affects morph types differently, we also break down performance by *root*, *derivational affix*, and *inflectional affix*. This follows the standard intuition behind borrowability scales: lexical material and derivational morphology tend to transfer more readily than inflectional endings, which are often more resistant in ordinary contact situations. In Czech in particular, inflectional endings are overwhelmingly native, so strong scores on inflectional morphs are expected.

Model	F1	Root	Deriv.	Infl.
All native	89.9	79.6	93.1	**100.0**
Memor.	94.4	87.7	97.0	**100.0**
Lemmatiz.	94.8	91.9	94.3	**100.0**
MLP30MC	96.5	92.1	97.6	**100.0**
GPT-4o-mini	90.9	81.3	93.5	**100.0**
GPT-5-mini	96.4	89.9	96.7	99.7
GPT-5.2	97.6	94.6	98.5	**100.0**
C-Opus4.5	**97.8**	**95.1**	**98.8**	**100.0**

Table 8: F1 by morph type on the test set. For LLMs we used the full training set in-context.

8. Conclusion and Future Work

We addressed the task of predicting the etymological origin of morphs in Czech, determining whether each morph is native or borrowed and, if borrowed, which languages are involved in its borrowing path. Because most resources focus on word-level etymology, there are currently no widely used datasets providing morph-level etymological annotation for segmented words. To fill this gap, we created a dataset of 300 Czech segmented sentences from the Shared Task on Morpheme Segmentation by SIGMORPHON (2022), containing 4,179 words and 10,245 morphs overall, of which 9,961 carry a non-empty etymological label. The resulting dataset is publicly available at `https://github.com/ampapacek/MorphemeOrigin`.

We compared three families of approaches: simple rule-based baselines, supervised classifiers, and LLM prompting with in-context examples. The supervised models use character n-grams together with structural cues (morph type/position and simple vowel-shape features). Among supervised systems, the strongest learning-based model was a multi-layer perceptron with a single hidden layer of 30 neurons, trained in a multi-label setup and augmented with *CzEtyL*-derived entries. It substantially outperformed the rule-based baselines. A bidirectional GRU over morph sequences was competitive, especially on borrowed morphs, but did not clearly surpass the simpler MLPs in our setting. We suspect that this is due both to the short morph sequences found inside words and to the relatively limited amount of labeled training data available for a higher-capacity sequence model. While the best supervised MLP does not match the very best in-context LLM setting, it remains attractive as a lightweight, deterministic, and inexpensive alternative that can be run locally without external API access or heavy computational resources. Semi-supervised self-training on the PDT corpus yielded no clear gains.

Overall, the best results come from LLMs when the full training set is provided in-context. In this setting, `Claude-Opus-4.5` reaches 97.8 % F1. Across LLMs, adding more in-context training morphs generally improves performance, especially for borrowed morphs.

Future work A straightforward next step is to enlarge the annotated dataset, since more examples expose models to a broader range of morphs and borrowing patterns. We also observed that scaling supervision helps: increasing in-context training improves LLM performance, and adding more labeled data benefits the supervised models as well. This may be particularly important for sequence-based models such as the GRU, whose advantages may only become clearer with larger annotated datasets. The annotation could be made more fine-grained, for example by distinguishing historical stages of Latin, German, or Greek, or by marking whether a morph is pan-Slavic or specific to Czech.

For learning models, richer shape-based features may help. For example, the vowel-start/end feature was useful, and encoding the full consonant–vowel (CV) pattern of each morph could capture additional regularities. On the model side, more structured architectures that predict borrowing paths directly, as well as ensembles combining complementary systems, could improve robustness. Another direction is to revisit semi-supervised methods such as self-training, which did not yield clear improvements in our current setup. Finally, it would be interesting to adapt pretrained models to this task (e.g., by fine-tuning) and to more systematically explore training and inference settings, including hyperparameters and prompting choices.

Although this study focused on Czech, the methodology is not language-specific and could be applied to other morphologically rich languages.

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9. Bibliographical References

Peter Arkadiev, Francesco Gardani, and Nino Amiridze. 2015. *Borrowed Morphology: An Overview*, pages 1–23. De Gruyter Mouton.

Mark Aronoff and Kirsten Fudeman. 2011. *What is Morphology?* Fundamentals of Linguistics. Wiley.

Khuyagbaatar Batsuren, Gábor Bella, Aryaman Arora, Viktor Martinovic, Kyle Gorman, Zdeněk Žabokrtský, Amarsanaa Ganbold, Šárka Dohnalová, Magda Ševčíková, Kateřina Pelegrinová, Fausto Giunchiglia, Ryan Cotterell, and Ekaterina Vylomova. 2022. The SIGMORPHON 2022 Shared Task on Morpheme Segmentation. In *19th SIGMORPHON Workshop on Computational Research in Phonetics, Phonology, and Morphology*, pages 103–116, Stroudsburg, PA, USA. Association for Computational Linguistics.

James P Blevins. 2016. *Word and paradigm morphology*. Oxford University Press.

Piotr Bojanowski, Edouard Grave, Armand Joulin, and Tomas Mikolov. 2017. Enriching word vectors with subword information. *Transactions of the Association for Computational Linguistics*, 5:135–146.

Philip Durkin. 2014. *Borrowed Words: A History of Loanwords in English*. Oxford University Press.

Thomas V. Gamkrelidze and Vjaceslav V. Ivanov. 1995. Chapter Four: The Structure of the Indo-European Root. In *Indo-European and the Indo-Europeans: A Reconstruction and Historical Analysis of a Proto-Language and Proto-Culture*, pages 185–187. De Gruyter Mouton, Berlin, Boston.

Francesco Gardani. 2008. *Borrowing of Inflectional Morphemes in Language Contact*. Phd thesis, University of Vienna.

Anthony P. Grant. 2020. *The Oxford Handbook of Language Contact*. Oxford University Press.

Edouard Grave, Piotr Bojanowski, Prakhar Gupta, Armand Joulin, and Tomas Mikolov. 2018. Learning word vectors for 157 languages. In *Proceedings of the International Conference on Language Resources and Evaluation (LREC 2018)*.

Martin Haspelmath. 2020. The morph as a minimal linguistic form. *Morphology*, 30(2):117–134.

Lars Johanson and Martine Robbeets. 2013. Copies versus cognates in bound morphology. *STUF – Language Typology and Universals*, 66(2):130–135.

Vojtěch John. 2024. Morph classifier. Master's thesis, Charles University, Faculty of Mathematics and Physics, Institute of Formal and Applied Linguistics (ÚFAL), Prague, Czech Republic.

Viktória Kovácsová. 2025. Automatic detection of lexical borrowings. Master's thesis, Charles University, Faculty of Mathematics and Physics, Institute of Formal and Applied Linguistics (ÚFAL).

LT4HALA Organizers. 2026. 4th workshop on language technologies for historical and ancient languages (lt4hala). `https://circse.github.io/LT4HALA/2026/`. Co-located with LREC 2026, Spain.

Oxford English Dictionary. 2025. school, n. `https://www.oed.com/dictionary/school_n1`. Oxford University Press, accessed 13 Oct 2025.

Aleš Manuel Papáček. 2025. Identification of morpheme origin.

Jiří Rejzek. 2019. *Český etymologický slovník*. 3rd edition. LEDA.

Frank Seifart. 2015. Direct and indirect affix borrowing. *Language*, 91(3):511–532.

Wiktionary. 2025. Wiktionary: The free dictionary. `https://www.wiktionary.org/`. Accessed: 2025-10-04.

Winston Wu and David Yarowsky. 2020. Computational etymology and word emergence. In *Proceedings of the Twelfth Language Resources and Evaluation Conference*, pages 3252–3259, Marseille, France. European Language Resources Association.

10. Language Resource References

Colin Goldberg. 2019. Morphemes dataset. `https://github.com/colingoldberg/morphemes`. Accessed: 2025-10-13.

Jan Hajič, Eduard Bejček, Jaroslava Hlaváčová, Marie Mikulová, Milan Straka, Jan Štěpánek, and Barbora Štěpánková. 2020. Prague Dependency Treebank – Consolidated 1.0. In *Proceedings of the 12th Conference on Language Resources and Evaluation (LREC 2020)*, pages 5208–5218, Marseille, France.

Vojtěch John, Zdeněk Žabokrtský, Benjamin Reeves, Magda Ševčíková, Haridanmu Abdukerim, Abduhalik Abduwaiti, Abdukerim Abliz, Ebrahim Ansari, Timofey Arkhangelskiy, Lizaveta Astapenka, Niyati Bafna, Khuyagbaatar Batsuren, Gábor Bella, Pier Marco Bertinetto, Chiara Celata, Hélène Deacon, Konstantinos Diamantopoulos, Šárka Dohnalová, Alexei Fedorenko, Matea Filko, Antonín Forst, Federica Gamba, Timur Garipov, Tanja Gaustad, Fausto Giunchiglia, Anna Glazkova, Hamid Haghdoost, Nabil Hathout, Irina Khomchenkova, Victoria Khurshudyan, Maria Klyucheva, Lukáš Kyjánek, Eleonora Maria Litta, Olga Lyashevskaya, Joël Macoir, Hugo Mailhot, Sun Maosong, Cindy McKellar, Maria Medvedeva, Dmitry Morozov, S.N. Muralikrishna, Fiametta Namer, Anna Nedoluzhko, Mahshid Nikravesh, Aleš Manuel Papáček, Marco Passarotti, Alexey Polyakov, Mihail Potapov, Mishra Pruthwik, Chrysanthi Raftopoulou, Ashwath Rao, Husain Samar, Claudia Sánchez-Gutiérrez, Iurii Savelev-Galiaminskii, Dipti Misra Sharma, Eleonora Slavíčková, Abishek Stephen, Emil Svoboda, Krešimir Šojat, Vanja Štefanec, Luigi Talamo, Jonáš Vidra, Arseniy Vydrin, Maximiliano A. Wilson, Liu Yang, and Aigul Zakirova. 2026. Universal segmentations 1.5 (UniSegments 1.5). LINDAT/CLARIAH-CZ digital library at the Institute of Formal and Applied Linguistics (ÚFAL).

Michal Olbrich, Viktória Brezinová, Šárka Dohnalová, Vojtěch John, Lukáš Kyjánek, Aleš Papáček, Emil Svoboda, Magda Ševčíková, Jonáš Vidra, and Zdeněk Žabokrtský. 2025. DeriNet 2.3. LINDAT/CLARIAH-CZ digital library at the Institute of Formal and Applied Linguistics (ÚFAL), Faculty of Mathematics and Physics, Charles University.

Jiří Rejzek, Aleš Papáček, Viktória Brezinová, and Zdeněk Žabokrtský. 2025. Czech Etymological Lexicon 1.0. LINDAT/CLARIAH-CZ digital library at the Institute of Formal and Applied Linguistics (ÚFAL), Faculty of Mathematics and Physics, Charles University.

SIGMORPHON. 2022. SIGMORPHON 2022 Shared Task on Morpheme Segmentation. `https://github.com/sigmorphon/2022SegmentationST`. Accessed: 2025-10-04.

Jana Straková, Milan Straka, and Jan Hajič. 2014. Open-Source Tools for Morphology, Lemmatization, POS Tagging and Named Entity Recognition. In *Proceedings of 52nd Annual Meeting of the Association for Computational Linguistics: System Demonstrations*, pages 13–18, Baltimore, Maryland. Association for Computational Linguistics.

A. Etymological Analysis of a Diachronic Czech Legal Corpus

In this appendix, we apply the best lightweight supervised system described in the main paper to a diachronic corpus of Czech legal texts and examine how the predicted proportions of native and borrowed morphs change over time. The goal of this analysis is not to provide a definitive historical account, but rather to illustrate one possible downstream use of the proposed morph-level etymology framework on a long real-world time series.

The corpus was constructed from publicly available Czech legislative texts obtained through the e-Sbírka API. We downloaded texts year by year and constructed one sampled file for each year by randomly sampling sentences up to approximately 1,000 words. The resulting dataset contains 109,257 words. The corpus was created within the CZDEMOS4AI project. Because it uses a fixed yearly word budget and includes only the texts available at the time of collection, it should be interpreted as a sample of legal language in each year rather than as a complete picture of legal language over the full period.

We then automatically segmented the corpus and applied the classifier from the main paper to the resulting morphs. For this analysis, we used the same MLP configuration as in our best lightweight system: a single hidden layer of size 30, multi-label prediction, and extended training data derived from the etymological dictionary. On the original test set in the main paper, this model achieves 96.5% F1.

The appendix should be interpreted cautiously for two reasons. First, the legal corpus belongs to a different domain than the data used for evaluation in the main paper, so the model's accuracy on legal texts may be lower. Second, both the morphological segmentation and classification were produced automatically, which means that some errors may already arise in the preprocessing stage. The figures and observations presented below should therefore be interpreted as exploratory rather than definitive.

In Table 9, we summarize the main predicted origin categories and the most frequent borrowed source languages. All shares are computed with respect to morphs rather than labels: the rows *Native* and *Borrowed* show the proportion of morphs assigned to these two broad classes, while the language-specific rows show the proportion of morphs whose predicted borrowing path includes the given language. Native morphs dominate overall, while Latin is the most prominent borrowed source language in every available year of the sample. Other prominent borrowed sources include English, German, French, and Greek, but all of them remain clearly below Latin in the aggregate picture.

Origin	Mean	Median	Max	Std. dev.
Native	91.04	91.22	94.61	1.95
Borrowed	8.96	8.78	15.25	1.95
Latin	5.84	5.55	10.78	1.61
English	1.74	1.69	3.42	0.43
German	1.47	1.38	2.72	0.50
French	1.19	1.13	3.02	0.48
Greek	1.16	1.05	3.97	0.58
Polish	0.34	0.32	1.67	0.20
Italian	0.23	0.18	0.93	0.19

Table 9: Summary of the main predicted origin categories and source languages across the years 1918–2026. Shares are computed with respect to all morphs within each year. For language-specific rows, a morph contributes to a language if that language appears anywhere in its predicted borrowing path. Standard deviation is computed across yearly shares. All values are in (%).

Across all sampled years, the mean predicted share of borrowed morphs is 8.96%. In the earliest available year, 1918, it is 6.03%, while the maximum in the current sample is observed in 2015, where it reaches 15.25%. Figure 2 shows the five-year weighted smoothing[4] of the overall borrowed-morph share together with fitted linear trend lines for that overall share and for the three most prominent borrowed source languages. Overall, the fitted trend line suggests a slight upward trend in the proportion of borrowed morphs over time, corresponding to roughly 2.5 percentage points per century.

The code, data, and generated outputs for this appendix are available at: `https://github.com/ampapacek/MorphemeOrigin/tree/main/legal_corpus_analysis`.

[4]Each year's value is replaced by a weighted average over a five-year window centered on that year, using weights 0.1, 0.2, 0.4, 0.2, and 0.1.

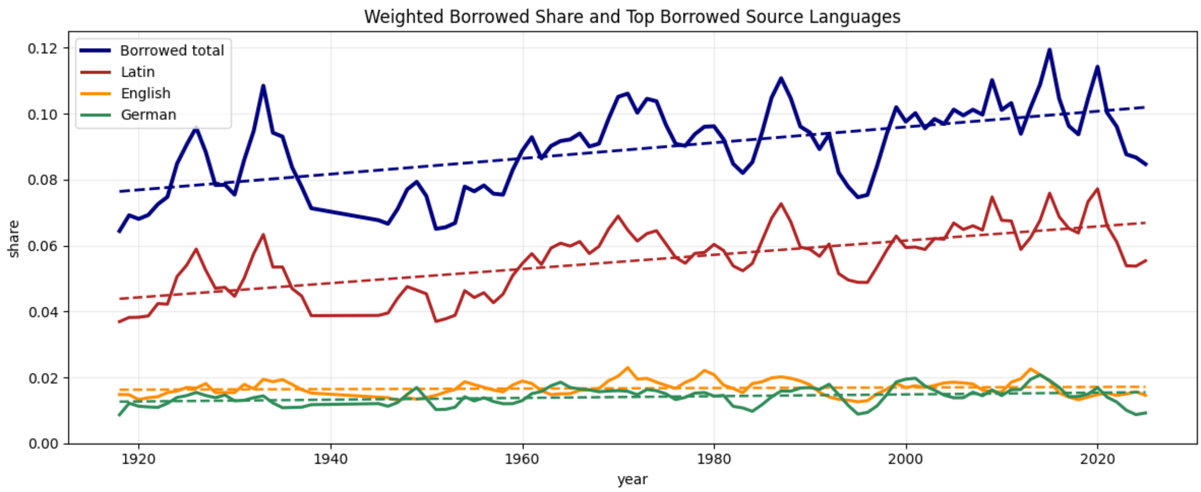

Figure 2: Five-year weighted smoothing of the overall predicted borrowed-morph share together with fitted linear trend lines for the overall borrowed share and for the three most prominent borrowed source languages: Latin, English, and German.